Fuzzy Factory: Product Performance and Profitability Analysis

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Problem Statement

- Analyze overall product performance
- Identify products generating the highest revenue
- Identify products contributing the highest profit
- Evaluate sales trends across all products
- Measure product-level profitability
- Study cross-product purchasing behavior (products bought together)
- Support better business decisions using data-driven insights
- Improve inventory planning based on performance and buying patterns

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About the Data

This project uses a Fuzzy Factory sales dataset from Maven Analytics, containing real e-commerce transaction records.
Three tables were used:

Orders – Order-level information and timestamps

Order items – Product-level sales data including price and cost

Products – Product details such as name and ID

These tables together support analysis of product performance, profitability, and buying patterns.

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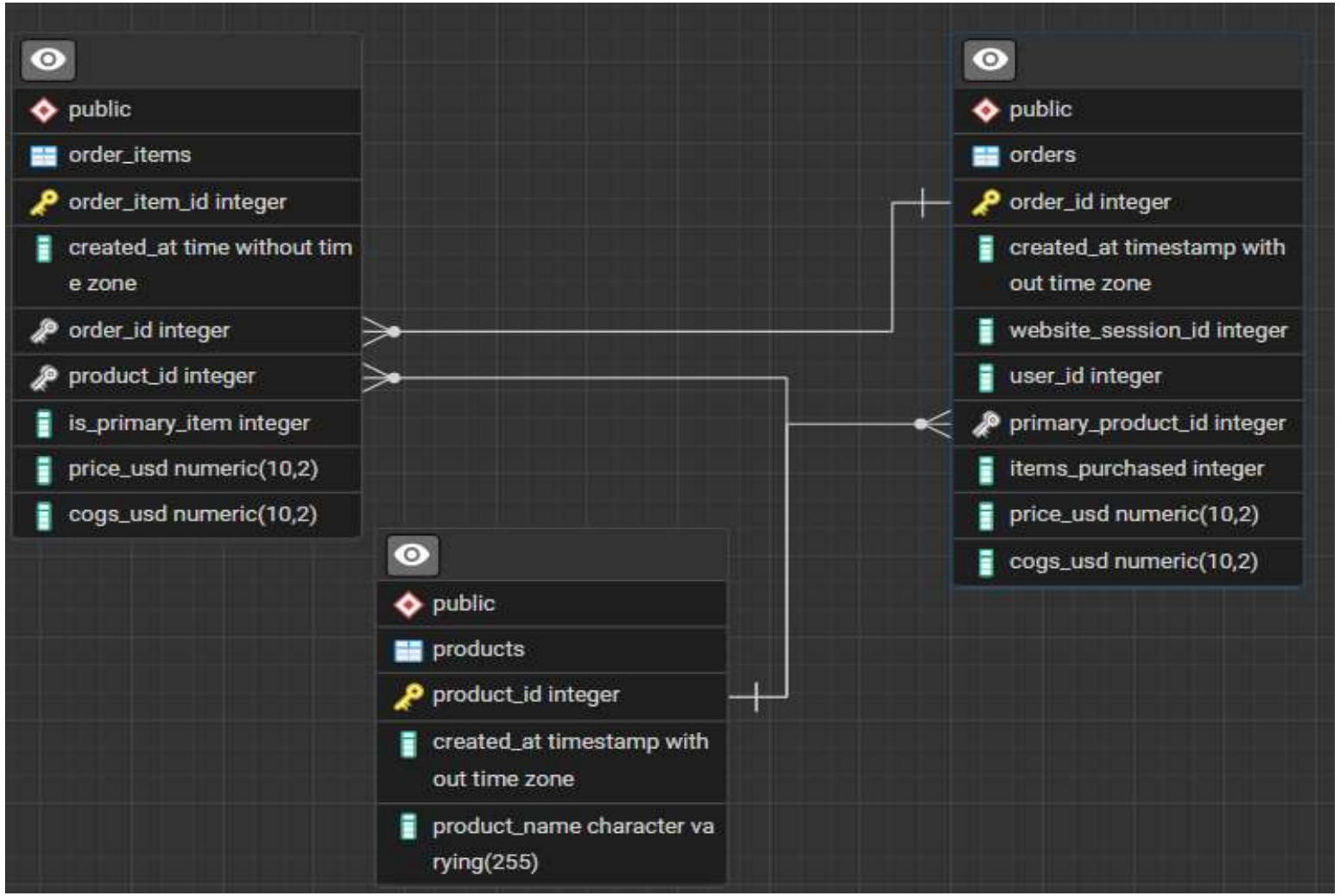
ER Diagram



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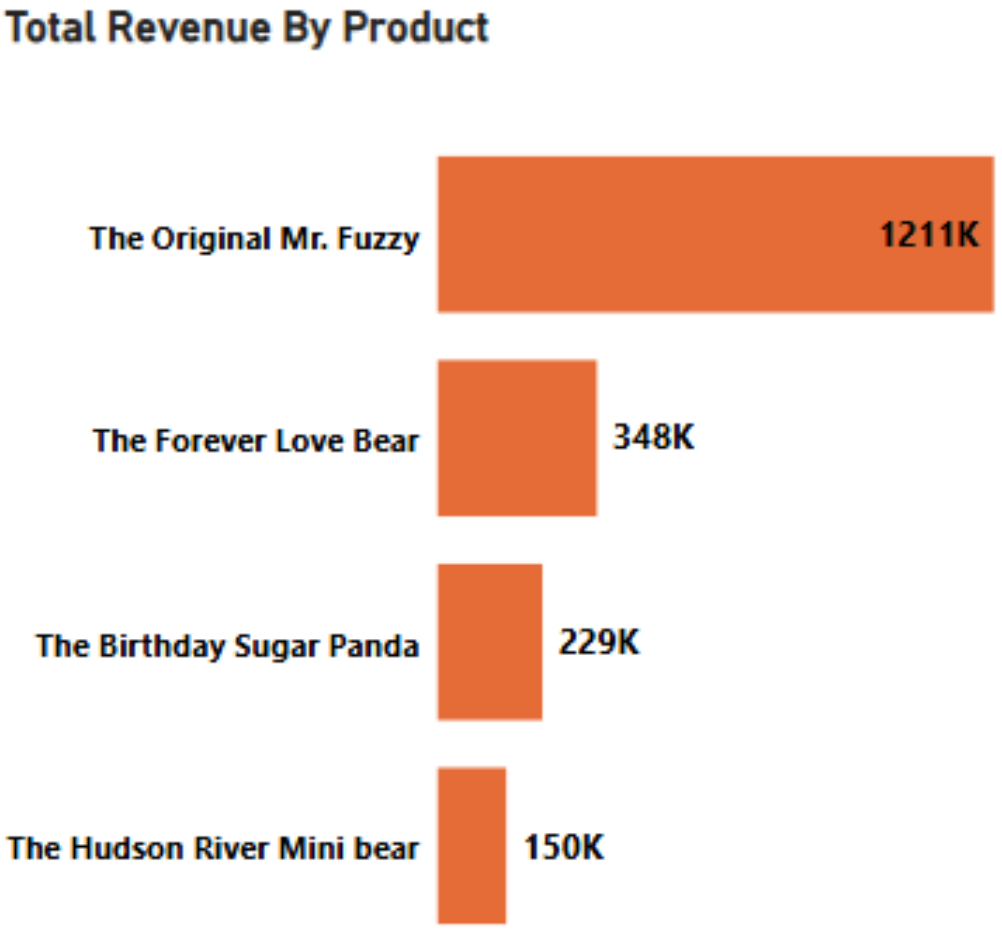
Proposed Solution

Q1:Total Revenue by Product Insights:

- The **Original Mr. Fuzzy** is the clear revenue leader, generating significantly **higher sales** than all other products.
- The **Hudson River Mini Bear shows** the **lowest revenue** contribution.
- This indicates that **Mr. Fuzzy** is the **primary** revenue driver for the business.



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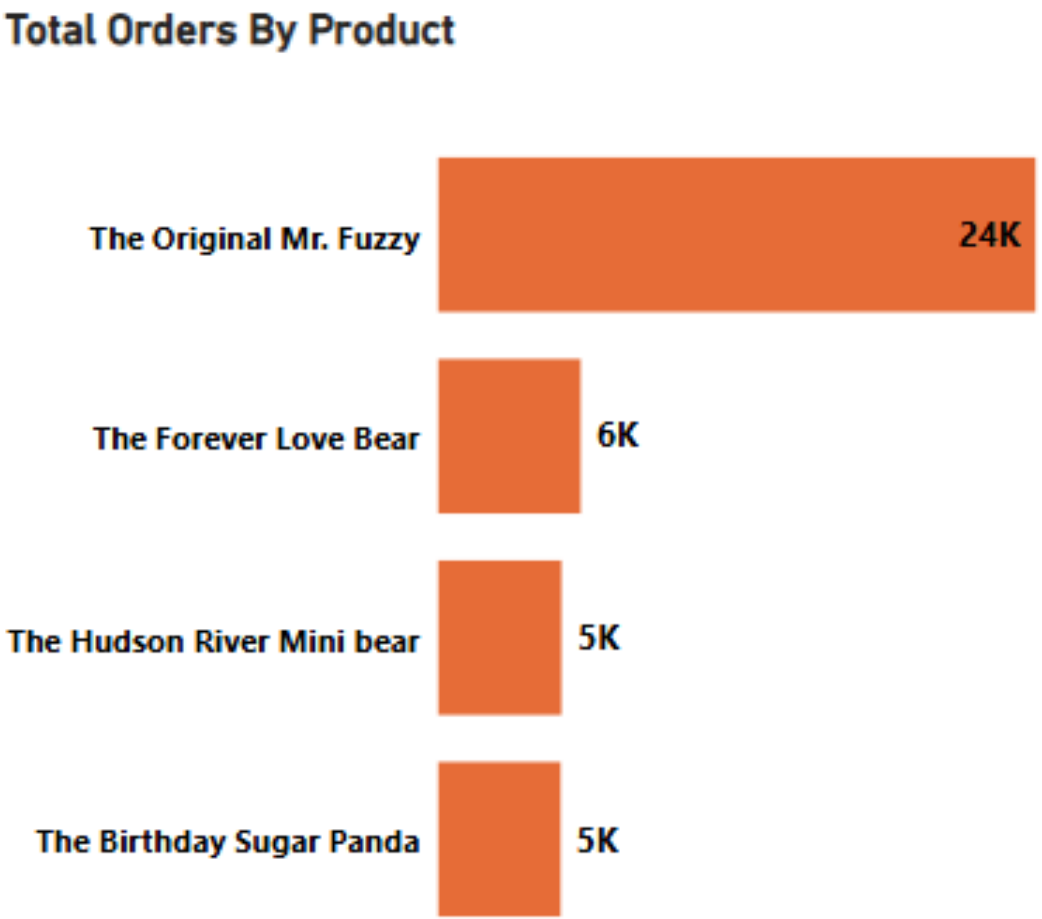
Proposed Solution

Q2:Total Orders by Product

Insight:

- **Mr. Fuzzy** also leads in total order volume, receiving **more customer** orders than any other product.
- The **Birthday Sugar Panda** records the lowest order count.



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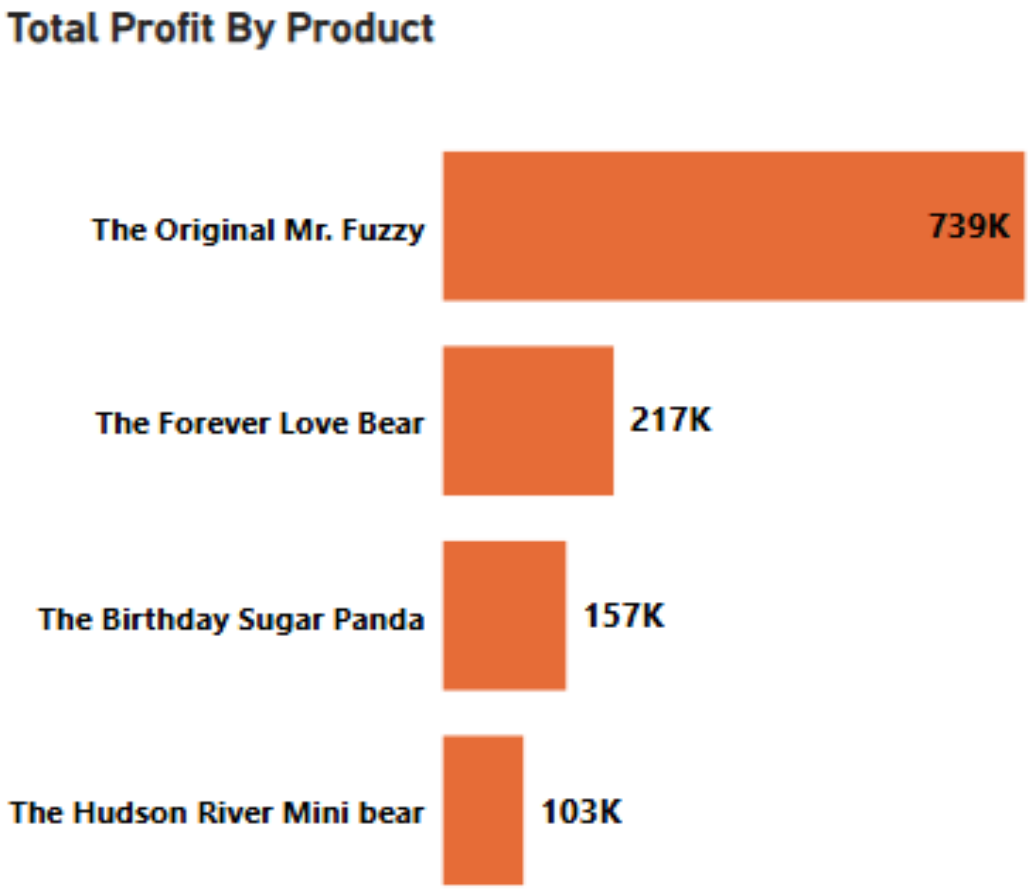
Proposed Solution

Q3:Total Profit by Product
Insight:

- **Mr. Fuzzy** generates the highest total profit.
- The Hudson River Mini Bear contributes the least profit among all products.
- Profit trends mirror revenue performance, showing that Mr. Fuzzy is both the highest-selling and the most profitable product.



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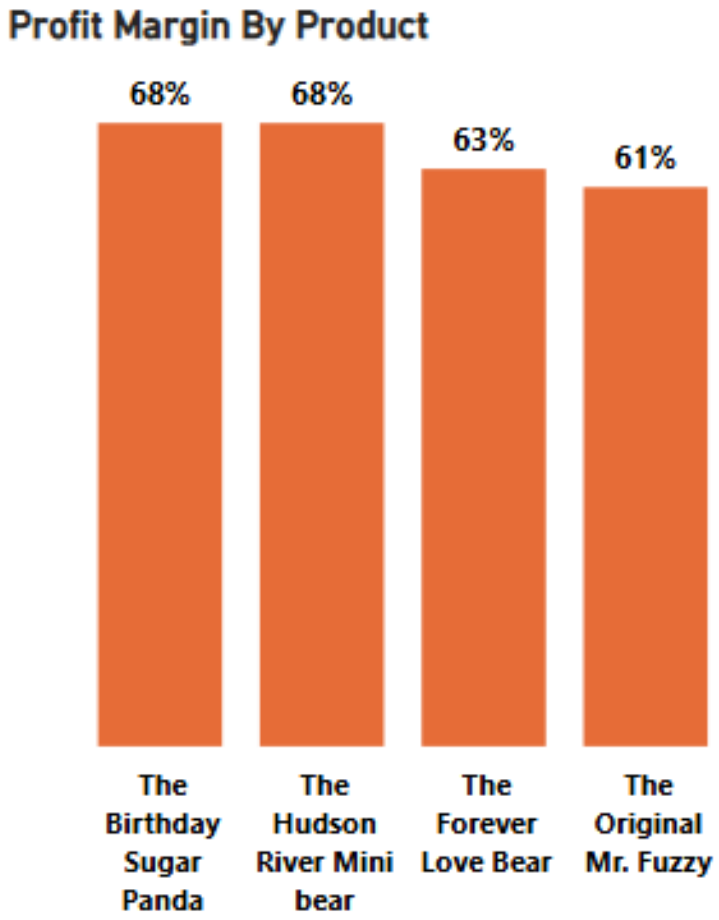
Proposed Solution

Q4: Profit Margin Analysis Insights:

- The **Birthday Sugar Panda** delivers the **highest profit margin**, indicating strong cost-to-revenue efficiency.
- In contrast, the **Original Mr. Fuzzy**, although the top seller by revenue and order count, shows the **lowest profit margin**, suggesting higher relative costs or lower pricing efficiency.



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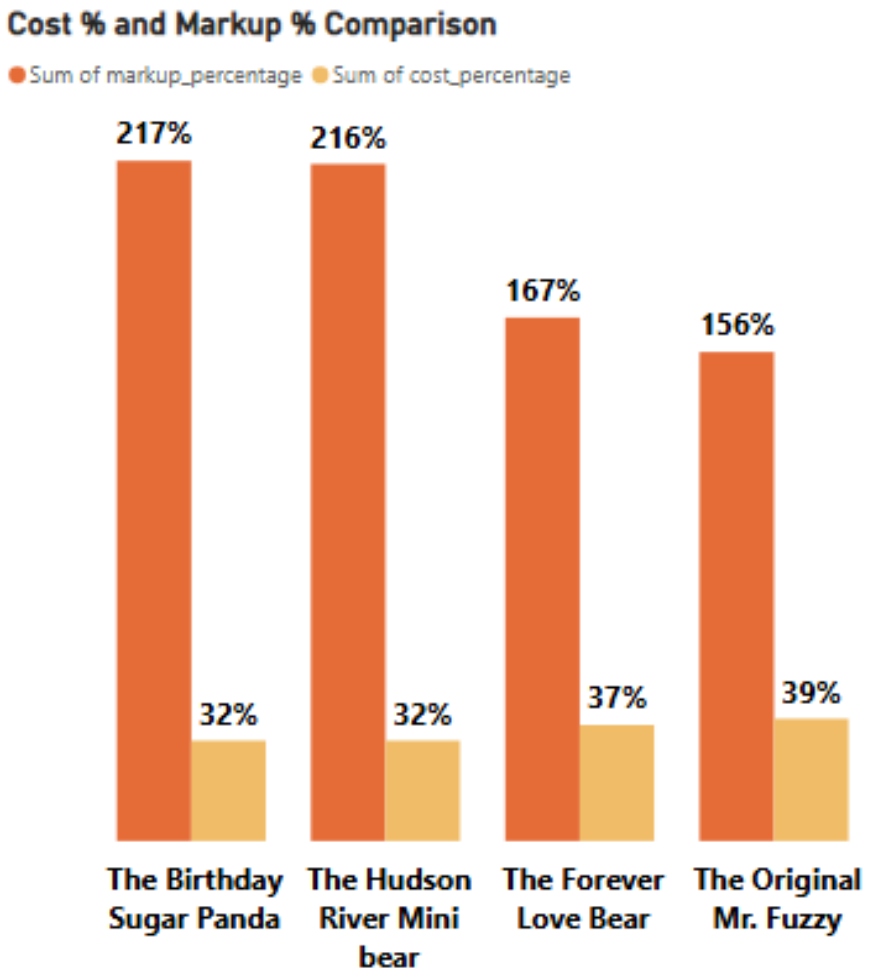
Proposed Solution

Q5: Cost % and Markup % Analysis Insight:

- Cost and markup patterns reveal clear differences in product efficiency.
  The **Original Mr. Fuzzy** carries the **highest cost percentage**, meaning it consumes a larger portion of cost per sale, while the **Hudson River Mini Bear** has the **lowest cost percentage**.

- For markup percentage, the **Birthday Sugar Panda** achieves the **highest markup**, making it the most pricing-efficient product, while the **Original Mr. Fuzzy** records the **lowest markup percentage**.



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Proposed Solution

Q6: Hero Product KPI Insight:

- The KPI comparison ranks each product across revenue, profit, and markup.

- The **Original Mr. Fuzzy** emerges as the overall **Hero Product**, performing consistently well across core business metrics.

Evaluation Of Hero Product

product_id	product_name	total_revenue	total_profit	markup_percentage	revenue_rank	profit_rank	markup_rank	hero_rank
1	The Original Mr. Fuzzy	1211058	738893	156%	1	1	4	6
2	The Forever Love Bear	347702	217350	167%	2	2	3	7
3	The Birthday Sugar Panda	229260	157028	217%	3	3	1	7
4	The Hudson River Mini bear	150490	102869	216%	4	4	2	10
Total		1938510	1216140	756%	10	10	10	30

- On the other end, the **Hudson River Mini Bear** acts as the **bleeding product**, showing weak performance in revenue rank (4), profit rank (4), and only moderate performance in markup rank (2), signaling an area requiring strategic improvement.

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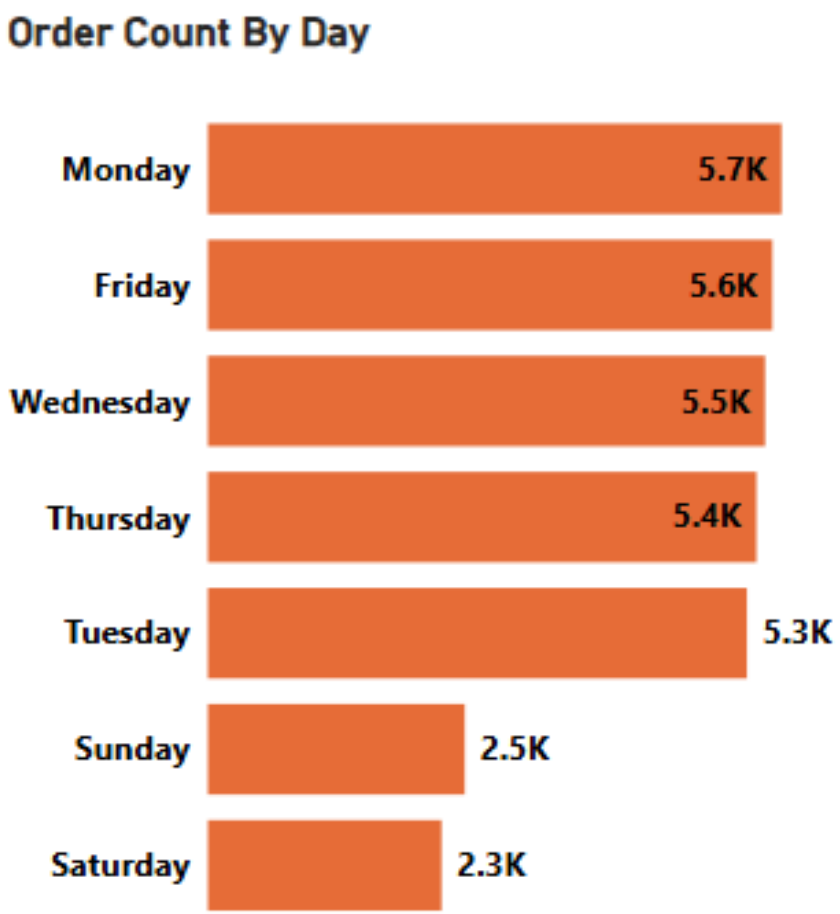
Proposed Solution

Q7:Day-Wise Order Frequency Insight:

- Sales activity varies across the week. **Monday records the highest number of orders**, indicating strong customer engagement at the start of the week.
- **Saturday shows the lowest order volume**, reflecting reduced weekend purchasing behavior.



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Proposed Solution

Q8 Product Co-Occurrence Analysis Insight

- Customers often purchase certain products together.
  The **most frequent combination** is:
  The Original Mr. Fuzzy + The Hudson River Mini Bear
  This pair shows a strong co-purchase relationship.

- The **least frequent combination**:
  The Forever Love Bear + The Birthday Sugar Panda
  This indicates weaker complementary demand between these products.

Product Co-Purchase

product_a	product_b	Sum of times_bought_together
The Original Mr. Fuzzy	The Hudson River Mini bear	3142
The Original Mr. Fuzzy	The Birthday Sugar Panda	2036
The Original Mr. Fuzzy	The Forever Love Bear	944
The Forever Love Bear	The Hudson River Mini bear	680
The Birthday Sugar Panda	The Hudson River Mini bear	662
The Forever Love Bear	The Birthday Sugar Panda	248
Total		7712

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Conclusion

Product Revenue & Profit Performance

- **Original Mr. Fuzzy** leads in total revenue, orders, and profit, making it the strongest overall performer.
- **Hudson River Mini Bear** consistently ranks the lowest across these sales metrics.
- Sales volume does not always reflect profitability — high-selling products can still show weaker margins.

Product Efficiency & Profitability Insights

- **Birthday Sugar Panda** and **Hudson River Mini Bear** deliver the **highest profit margins and markup percentages**, indicating strong cost efficiency.
- The **Hero Product KPI** identifies Original Mr. Fuzzy as the most balanced product across revenue, profit, and markup.

Customer Behaviors & Product Pairing

- **Monday is the highest-order day**, while **Saturday records the lowest**, showing clear weekday strength.
- The strongest co-occurring product pair is **Original Mr. Fuzzy + Hudson River Mini Bear**, highlighting strong cross-selling potential.

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Future Scope

Investigate Low Sales of High-Profit Products

- Products with strong profitability—such as *Birthday Sugar Panda* and *Hudson River Mini Bear*—are not generating high sales.
- The business should explore **why these efficient products are underperforming** (pricing, visibility, customer preference, positioning, etc.).

Operational Planning Around Peak Order Days

- Since **Monday shows the highest order volume**, inventory and staffing should be optimized for this day
- Avoid scheduling employee leave during high-traffic weekdays to maintain service efficiency.

Optimize Product Bundling Strategies

- The most frequent bundle is **Original Mr. Fuzzy + Hudson River Mini Bear**.
- The business can leverage this by creating **bundle offers, promotions, or cross-selling strategies**, while also improving visibility of low-pairing combinations.

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THANK YOU

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